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Deep Learning for Bidirectional Translation between Molecular Structures and Vibrational Spectra

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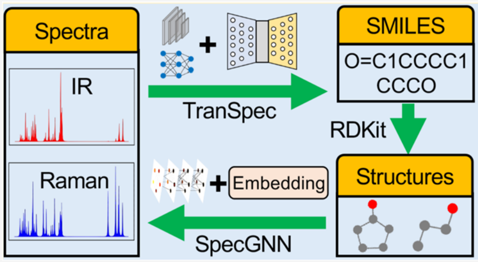

ABSTRACT: Two deep learning models, TranSpec and SpecGNN, were developed to establish a bidirectional mapping between molecular vibrational spectra and simplified molecular input line entry system (SMILES) representations, akin to a "translation" between the language of spectra and the language of molecular structures. Initially, TranSpec achieved accuracy rates of 55 and 63% for quantum chemistry (QC)-calculated IR and Raman spectral data sets, respectively, but its performance dropped to 11% for the NIST experimental IR data set. To address this, we combined IR and Raman spectra as input; augmented the data set; employed model fusion, transfer learning, and multisource learning; applied molecular mass filtering; and leveraged SpecGNN for spectral simulation and candidate reordering. These improvements boosted TranSpec's accuracy to 53.6% for the experimental IR data set. Notably, SpecGNN outperformed traditional QC methods in terms of both spectral accuracy and computational efficiency. Finally, we demonstrated TranSpec's ability to recognize functional groups and distinguish isomers or homologues. Together, TranSpec and SpecGNN models provide an efficient and accurate AI-driven framework for interpreting molecular structures and spectra, advancing applications in spectroscopy and cheminformatics.

1. INTRODUCTION

Vibrational spectroscopy, including IR and Raman spectroscopy, plays a crucial role in identifying chemical compounds[1,2] due to its high sensitivity, its nondestructivity, and its ability to provide molecular fingerprints. In practical applications, chemical compounds are often identified using spectral library matching,[3,4] which compares the query spectrum with those in a reference library. This method is highly efficient and accurate in domains with well-defined chemical libraries, such as astrochemistry or specialized chemical engineering processes.[5−7] However, its reliance on predefined data sets limits its ability to identify novel compounds, predict their unknown properties, or enable inverse molecular design, thereby constraining its applicability in exploratory research and molecular innovation.

As another common approach, the "trial-and-error" method[8] adopts a cyclic, self-consistent workflow involving the construction of three-dimensional molecular structures, spectral simulation, comparison with the query spectrum, and adjustment of the molecular model. However, this process is highly complex, requiring deep chemical intuition, extensive experience, and costly calculations, which largely limit its efficiency and accuracy.

Due to the superiority of machine learning (ML) at finding hidden correlations, it has been exploited in library matching and "trial-and-error" methods.[9−12] For example, Zhang et al. developed a deep-learning-based neural network to predict molecular fingerprints from electron ionization−mass spectrometry and to search molecular structure data sets with the predicted fingerprints.[13] Similarly, Wang et al. trained a contrastive learning network to match ^{13}C NMR spectra directly against a molecular structure library using SMILES representations.[14] While these approaches partially address the low coverage rates of spectral data sets by transforming spectral data into searchable indices or cross-referencing structural libraries, their identification accuracies and efficiencies remain limited. Another strategy involves using ML-assisted molecular generation to replace computational molecular design, significantly improving the efficiency of "trial-and-error" methods. For example, Tsuda et al. proposed a de novo molecular generator,[15] and Priyakumar et al. employed the Monte Carlo tree search (MCTS) and a message passing neural network to generate the molecules.[16] Molecular recognition is then achieved by comparing the NMR spectra of the generated molecules with the query spectra. However, this process still relies on quantum chemistry calculations, which significantly limits its efficiency.

Meanwhile, ML has also been widely applied in spectral simulations. For example, Dral and co-workers proposed a universal ML model from the AIQM series, designed to

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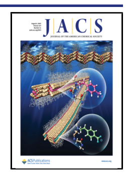

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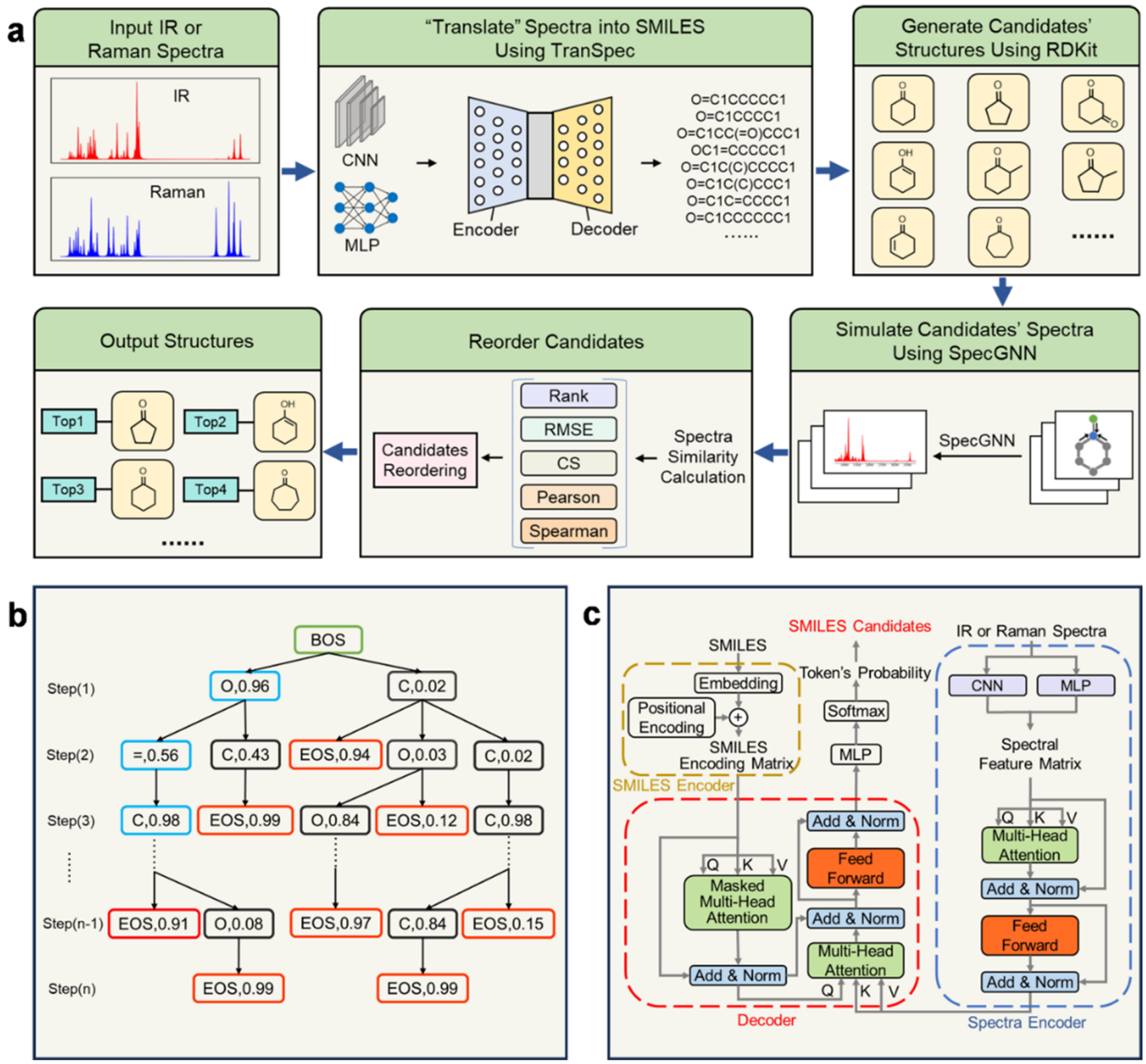

Figure 1. Workflow of molecule recognition using TranSpec and SpecGNN. (a) Schematic diagram of the workflow: TranSpec first translates IR or Raman spectra into candidate SMILES representations, and SpecGNN simulates the spectra of these candidates and reorders them based on spectral similarity. (b) Illustration of TranSpec predicting tokens step-by-step, along with their corresponding probabilities. (c) Schematic diagram of TranSpec's architecture, with color-coded components highlighting its individual elements.

approach the accuracy of CCSD(T)/CBS for molecular IR spectra.[17−19] It achieves accuracy comparable to density functional theory (DFT) (compared to the experiment) while operating at the speed of the semiempirical GFN2-xTB method.[17] Ghosh et al. used multilayer perceptron (MLP) and convolutional neural networks (CNNs) to predict electronic excitation spectra.[20] Additionally, Müller et al. developed FieldSchNet,[21] Jiang et al. developed FIREANN,[22] and Gastegger et al. developed PaiNN[23]—methods that enable molecular spectral simulations based on ML-driven molecular dynamics (MD) simulations. However, these MD-based methods remain computationally expensive, making them impractical for high-throughput screening applications. From this perspective, developing more efficient and accurate ML methods for simulating molecular spectra is highly desirable.

The Transformer model[24] has demonstrated remarkable success in various fields such as image recognition, speech recognition, and natural language processing.[25−27] Its multi-head attention mechanism not only improves the translation accuracy of very long sentences but also processes words precisely and with contextually appropriate meanings. IR and Raman spectra contain abundant information about molecular structure, including details about specific functional groups, local environments, and nonlocal chemical environments. On the other hand, molecules can also be equivalently represented as text in line symbol format, such as the international chemical identifier (InChI),[28] simplified molecular input line entry system (SMILES),[29] and SMILES arbitrary target specification (SMARTS). As a result, an alternative approach to identify the

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unknown compounds is to map, or "translate", the spectral sequence to the widely accepted molecular representations.

In this work, we developed a deep learning method called TranSpec to directly translate IR or Raman spectra into SMILES representations and a graph neural network called SpecGNN to efficiently and accurately simulate the experimental molecular vibrational spectra. Using either a convolutional neural network[30−32] (CNN) or a multilayer perceptron[33−35] (MLP), we first extracted the spectral feature matrices from the IR and Raman spectra. These features were then used to generate possible tokens, output their corresponding probabilities sequentially, and produce the potential SMILES candidates. Next, we employed SpecGNN to simulate the spectra for all SMILES candidates generated by TranSpec and reordered the candidates based on spectral similarity to improve the accuracy of the translation process (illustrated in Figure 1a). Finally, we explored the performance of TranSpec in other tasks, such as identifying functional groups and distinguishing isomers and homologues.

2. EXPERIMENTAL DETAILS

Data Collection and Preprocessing. We first trained TranSpec using the harmonic vibrational spectra of the 130,000 organic molecules in the QM9S data set. Then, we downloaded 5624 experimental IR spectra in the gas phase from the NIST Chemistry WebBook to verify the TranSpec's transferability. For the sake of training, we unify all of the spectral data from the QM9S, QM14S, and experimental IR data set. For QM9S, we normalized the IR and Raman spectra that range from 0 to 4000 cm^{-1} with a resolution of 1.33 cm^{-1}, so that the maximum spectral intensity is equal to 100. We only considered the spectral intensities and ignored the wavenumbers, as the wavenumber range is consistent across all molecules. This process will generate a 1×3000 tensor for each individual spectrum. Packing IR and Raman spectra together generates a 2×3000 tensor. Considering the experimental IR spectral range from 552 to 3844 cm^{-1}, we extracted the intensity values in the same range for the QM9S and the QM14S data sets. Then, we set a resolution of 1.09 cm^{-1} for these spectra to produce a 1×3000 tensor, which will then be used in the transfer learning or multisource learning.

For SMILES representations, we used Kekuler instead of canonical style to reduce the number of labeled species. Moreover, all SMILES were normalized using the RDKit[36] to keep the uniformity of naming rules. Finally, we used 16 tokens to represent the start (⟨BOS⟩), termination (⟨EOS⟩), padding (⟨PAD⟩), two special chemical bonds (⟨=⟩, ⟨#⟩), four elements (⟨C⟩, ⟨N⟩, ⟨O⟩, ⟨F⟩), two branches (⟨(⟩, ⟨)⟩), and five kinds of rings or connections (⟨1⟩, ⟨2⟩, ⟨3⟩, ⟨4⟩, ⟨5⟩) that exist in the QM9S data set. Except for the above 16 kinds, the three elements (⟨Cl⟩, ⟨Br⟩, ⟨S⟩) were added for the experimental IR and QM14S data set. During the training, all of the data sets are operated separately and divided into training, test, and validation sets with proportions of 80, 10, and 10%, respectively. It is noted that all of the results in the present work are counted from the test set.

TranSpec Architecture. We build the encoder−decoder structure and utilize the multihead attention mechanism[37,38] to translate the molecular spectra into the corresponding SMILES. As illustrated in Figure 1c, TranSpec consists of spectral encoder, SMILES encoder, and decoder. In the spectral encoder, we first extracted the spectral feature matrices from IR or Raman spectra using either a convolutional neural network (CNN) or a multilayer perceptron (MLP). Here, we named the two models TranSpec-CNN and TranSpec-MLP for the sake of discussion. The architecture and detailed settings of the CNN and MLP are provided in "Architecture of CNN and MLP" in the Supporting Information. Then, we utilize the multihead self-attention mechanism[39] with 8 attentional heads to simultaneously mine the relationship among the different positions of the query spectral sequence. A feed-forward network[40] is used to introduce nonlinearities and enhance the model representation. Finally, the spectral encoder module produces the key matrix K^B and value matrix V^B that are used in the next decoder module. In the SMILES encoder, every token was alphabetically and positionally embedded into a $d_{model} = 512$ dimensional matrix, generating the SMILES encoding matrix.

In the decoder, we passed the SMILES encoding matrix through a masked multihead self-attention mechanism,[41] ensuring that the model could jointly attend to the information on different tokens in SMILES. In this process, the padding mask and look-ahead mask were utilized to ensure that the prediction of every token depends only on the already generated ones. After passing a residual connection[42] (labeled as "Add and Norm" on the left panel of the Decoder in Figure 1c), we produced the query matrix Q^A that is used in the next multihead attention mechanism. It is noted that Q^A represents the information on the tokens that need to be accurately translated from spectra, while K^B represents the corresponding information hidden in the spectra. On the other hand, V^B represents the whole information on the spectra. Then, combining the scaling and SoftMax[43] operation on Q^A, K^B, and V^B predicts the probability distributions for the next token. Here, the formula of the multihead attention mechanism could be written as

$$\text{multihead}(Q, K, V) = \text{concat}(\text{head}_1, \cdots \text{head}_h)W^{o} \tag{1}$$

$$\text{head}_i = \text{attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{2}$$

$$\text{attention}(Q^A, K^B, V^B) = \text{softmax}\left(\frac{Q^A(K^B)^T}{\sqrt{d_k}}\right)V^B \tag{3}$$

where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{model} \times d_q}$, $W_i^K \in \mathbb{R}^{d_{model} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{model} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{model}}$. In the present work, we employ $h = 8$ parallel attention heads, resulting $d_q = d_k = d_v = \frac{d_{model}}{h} = 64$.

Finally, we applied the MLP and SoftMax functions on the output of the decoder to generate the probability distribution of the current token. All parameters in the network are optimized by minimizing the cross-entropy loss[44] between the output and the target sequence

$$L(y, \hat{y}) = -\sum_{i=1}^{N} y_i \cdot \log(\hat{y}_i) \tag{4}$$

where y is the target label, $\hat{y}$ is the result predicted by the model, and N is the number of categories.

SpecGNN Architecture. To efficiently and accurately simulate the molecular experimental spectra, we constructed SpecGNN to translate SMILES into spectral sequences. As shown in Figure 3a, the SMILES was first transformed into a two-dimensional (2D) molecular graph using the RDKit toolkit, obtaining the binary feature representations for atom ($F_i^{atom} \in (0,1)$) and chemical bonds ($F_{ij}^{bond} \in (0,1)$). Then, the atomic and bond features were embedded into the initial node features h_i^0 and edge features e_{ij}^n

$$h_i^0 = W_1W_2(F_i^{atom}) \tag{5}$$

$$e_{ij}^n = W_1W_2(F_{ij}^{bond}) \tag{6}$$

where W_1 and W_2 are the trainable weight matrix. We used the graph isomorphism network with edge feature convolution (GINEConv)[45] to generate the interaction message between the node and edge features M_{ij}^n

$$M_{ij}^n = \text{MLP}\left((1 + \epsilon) \times h_i^n + \text{ReLU}\sum_j (h_i^n + e_{ij}^n)\right) \tag{7}$$

where ε is a learnable parameter. Then, the node feature update as

$$h_{ij}^{n+1} = \text{SiLU}(L_gL_l(M_{ij}^n)) \tag{8}$$

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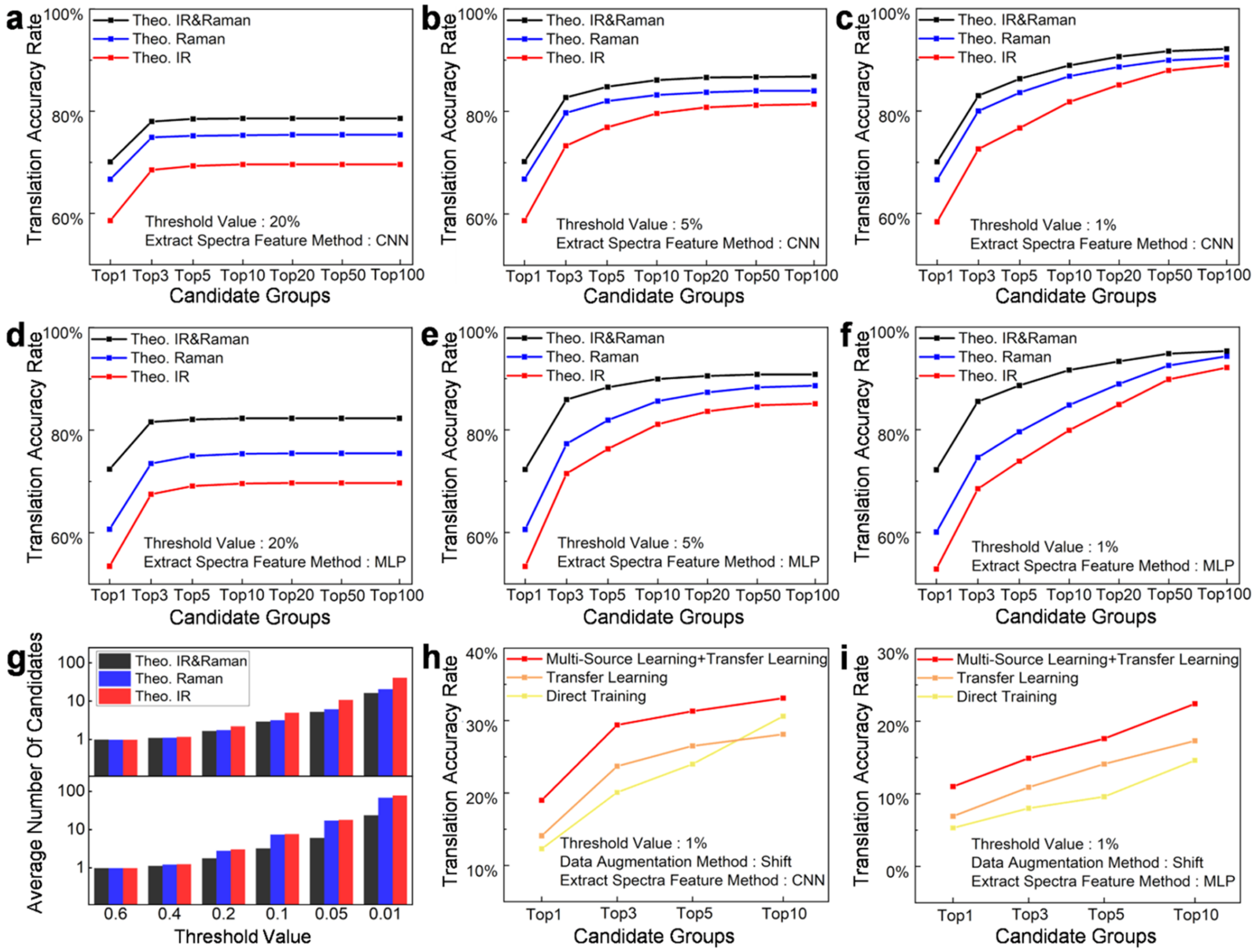

Figure 2. Translation accuracy of TranSpec. (a−c) Translation accuracy of TranSpec-CNN on the QM9S data set for various TopN candidate groups at threshold values of 20, 5, and 1%. (d−f) Similar results to panels (a−c) but using the TranSpec-MLP model. (g) Comparison of the average number of SMILES candidates produced by TranSpec-CNN and TranSpec-MLP on the QM9S data set with different threshold values. (h, i) Comparison of the translation accuracy on the experimental IR data set using transfer learning, multisource learning, and direct training.

where SiLU, L_l, and L_g represent the Sigmoid activation, layer normalization, and graph normalization, respectively. The two normalizations serve to slow over the smoothing of the GNN and prevent the gradient explosion. Finally, the spectral sequence of the whole molecule is predicted by averaging the corresponding contribution from the nodes through the MLP

$$\text{spec} = \text{mean}(\text{MLP}(h_i^N)) \tag{9}$$

A hybrid loss function that combines spectral information divergence (SID)[46] and root-mean-square error (RMSE) is used to optimize the parameters in SpecGNN

$$\text{loss} = a \times \text{SID}(p, q) + b \times \text{RMSE}(p, q) \tag{10}$$

$$\text{SID}(p, q) = \sum_{i=1}^{n} \left(p_i \log\frac{p_i}{q_i} + q_i \log\frac{q_i}{p_i} \right) \tag{11}$$

$$\text{RMSE}(p, q) = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (p_i - q_i)^2} \tag{12}$$

where a and b are the variable parameters, currently optimized to 0.001 and 1, respectively. $p = \{p_1, p_2, \cdots, p_n\}$ and $q = \{q_1, q_2, \cdots, q_n\}$ are the predicted and target spectral vectors, respectively.

3. RESULTS AND DISCUSSION

The Translation of Theoretical Spectra into SMILES Using TranSpec. We previously constructed the QM9S[47] data set, which includes not only the scalar, vectorial, and tensorial properties of molecules but also their IR and Raman spectra, simulated using the Gaussian 16 package[48] at the B3LYP/deft-tzvp level.[49,50] TranSpec was first trained using the theoretically simulated harmonic vibrational spectra of the 130k organic molecules in this data set. The optimized hyperparameters are listed in Table S1. As shown in Figure 1b, TranSpec outputs the tokens and their corresponding probabilities sequentially. To prevent an infinite number of output SMILES records, a threshold value for the probabilities must be set. Additionally, to reduce computational costs, the maximum number of SMILES candidates was limited to 500. The candidates were then ranked by multiplying the probabilities at each step until the termination token appeared. We define a “candidate group” as the set of the N most probable SMILES (referred to as TopN), and the translation is considered successful if the correct SMILES is included in the TopN candidates.

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Table 1. Improved Translation Accuracy Was Achieved Using Different Methods

approaches to increase the translation accuracy		data set	Top 1 (%) CNN	Top 1 (%) MLP	Top 3 (%) CNN	Top 3 (%) MLP	Top 5 (%) CNN	Top 5 (%) MLP	Top 10 (%) CNN	Top 10 (%) MLP
untreated		Theo. IR	58.4	52.9	72.6	68.5	76.7	73.9	81.8	79.9
		Theo. Raman	66.6	60.0	80.0	74.6	83.6	79.6	86.8	84.8
		Theo. IR and Raman	70.1	72.2	83.0	85.5	86.3	88.6	88.9	91.6
		Exp. IR	16.4	5.9	22.2	11.7	25.6	14.8	30.8	19.0
data augmentation methods	quantizing	Theo. IR	60.2	52.4	73.4	68.6	77.5	74.0	81.3	80.3
		Theo. Raman	72.0	60.2	83.7	74.3	86.3	79.4	88.4	84.8
		Theo. IR and Raman	73.2	72.3	85.1	85.7	87.8	88.6	90.2	91.7
		Exp. IR	19.8	9.6	29.4	15.1	31.0	17.4	33.5	21.5
	scaling	Theo. IR	61.6	53.8	75.5	69.9	79.1	75.1	82.4	81.0
		Theo. Raman	71.5	61.8	82.7	76.8	85.5	81.6	87.6	86.2
		Theo. IR and Raman	73.4	73.9	85.1	86.6	87.6	89.2	89.4	91.3
		Exp. IR	18.1	9.3	24.2	15.5	26.9	18.0	31.7	20.5
	shifting	Theo. IR	62.9	57.8	77.2	73.6	80.7	78.9	84.1	84.2
		Theo. Raman	74.1	63.9	86.1	78.1	88.9	83.0	90.9	87.7
		Theo. IR and Raman	75.3	74.5	87.1	87.7	89.7	90.2	91.8	92.2
		Exp. IR	19.0	11.0	29.4	14.9	31.3	17.6	33.1	22.4
model fusion	ensemble of eight sets of SMILES candidates[a]	Theo. IR	76.7		92.6		95.5		97.6	
		Theo. Raman	83.3		95.5		97.5		98.5	
		Theo. IR and Raman	87.6		97.4		98.5		99.3	
		Exp. IR	23.5		38.3		45.0		51.2	
	ensemble of 16 sets of SMILES candidates[b]	Exp. IR	25.1		39.7		45.7		53.6	
filtering candidates with molecular mass		Exp. IR	49.8		64.6		67.6		68.3	
reordering candidates using SpecGNN		Exp. IR	32.4		47.5		53.6		59.1	
combined reordering using SpecGNN and filtering with mass		Exp. IR	53.6		66.0		67.6		68.5	

[a]Eight sets of SMILES candidates generated from TranSpec-CNN and TranSpec-MLP models trained on quantized, scaled, shifted, and untreated data sets. [b]Sixteen sets of SMILES candidates produced by varying model parameters to create two versions of TranSpec-CNN and two versions of TranSpec-MLP, each trained on quantized, scaled, shifted, and untreated data sets.

Two versions of TranSpec were developed: TranSpec-CNN, which uses a CNN for feature extraction, and TranSpec-MLP, which utilizes an MLP for feature extraction. Figure 2a−f shows the translation accuracies on the QM9S data set using TranSpec-CNN and TranSpec-MLP under different settings. The Top 1 translation accuracy for TranSpec-CNN is consistently approximately 58.4, 66.6, and 70.1% for the IR, Raman, and IR and Raman spectra, respectively, regardless of the threshold value set. Higher translation accuracy can be achieved by including more candidates, as seen for Top 3, Top 5, or Top 10. Smaller threshold values, rather than larger ones, favor increased translation accuracy. This can be attributed to the fact that smaller threshold values retain tokens with very low probabilities, even though the computational cost increases exponentially (Figure 2g). Additionally, the translation accuracy from Raman spectra is consistently higher than that from IR spectra, and combining IR and Raman spectra further improves translation accuracy by 3−5%.

While the Top 1 translation accuracy of TranSpec-MLP is lower than that of TranSpec-CNN when using a single IR or Raman spectrum, this trend reverses when IR and Raman spectra are used together, with TranSpec-MLP outperforming TranSpec-CNN. This phenomenon suggests that TranSpec-CNN excels at capturing localized features within individual IR or Raman spectra, while TranSpec-MLP is more effective at leveraging the correlation and interactions between the IR and Raman spectra.

To improve its translation accuracy, we incorporated data augmentation and model fusion into TranSpec. First, we applied quantizing, scaling,[51] and shifting[52] to artificially introduce noise into the spectra. As shown in Table 1, these three data augmentation techniques improved translation accuracy by 2−5%, with shifting providing the biggest improvement. For model fusion, we combined all SMILES candidates generated by two different models (TranSpec-CNN and TranSpec-MLP) trained on distinct artificially modified spectra (quantized, scaled, shifted, and untreated data sets). These candidates were then reranked using a voting mechanism[53] to ensure the correct SMILES achieved a higher ranking. As described in Table 1, model fusion improved translation accuracy on the QM9S data set by 9−13%. Finally, the Top 1 and Top 10 translation accuracies using IR and Raman spectra reached 87.6 and 99.3%, respectively.

TranSpec's Generalizability to Experimental Spectra. We evaluated the transferability of TranSpec using the 5624 experimental IR spectra extracted from the NIST Chemistry WebBook.[54] To ensure data uniformity, we used only gas-phase spectra. The threshold value was uniformly set to 1% based on training results from the QM9S data set. As shown in Figure 2h−i, direct training on experimental IR spectra using TranSpec-CNN and TranSpec-MLP resulted in low translation

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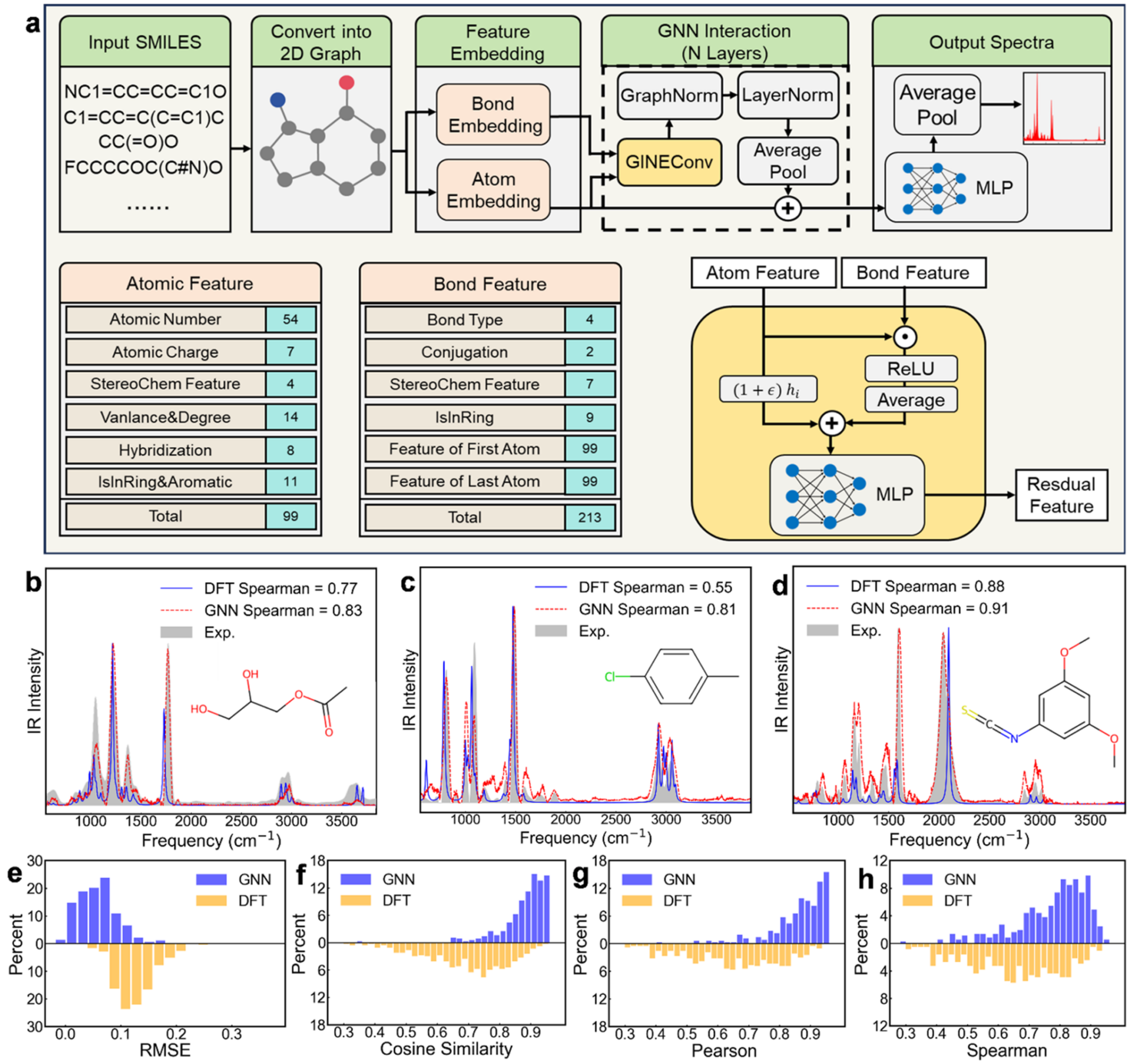

Figure 3. Architecture of SpecGNN and its performance in spectral simulations. (a) Architecture of the SpecGNN graph neural network for simulating experimental IR spectra based on SMILES representations, with color-coded components highlighting individual elements of the model. (b−d) Comparison of the SpecGNN-predicted, DFT-predicted, and experimental spectra for three molecules, 2,3-dihydroxypropyl formate, 4-chlorotoluene, and 3,5-dimethoxyphenyl isothiocyanate, randomly selected from the test set of the experimental IR data set. (e−h) Evaluation metrics, including RMSE, cosine similarity, Pearson correlation coefficient, and Spearman correlation coefficient, comparing SpecGNN-predicted and DFT-predicted spectra against the experimental spectral benchmark.

accuracies of 12.3 and 5.3%, respectively. This can be attributed to the small size of the experimental IR data set.

Transfer learning,[55] multisource learning,[56] and model fusion[57] were employed to enhance the translation accuracy. Given that the experimental IR data set includes additional elements (Cl, Br, and S) not present in the QM9S data sets, we randomly selected 20,000 molecules from PubChem[58] with a matching elemental composition and conducted QC calculations to generate their IR spectra. These molecules, which contain up to 14 heavy atoms, constitute the QM14S data set.[59] The QM9S and QM14S data sets were combined to pretrain TranSpec, followed by fine-tuning on the experimental data set. This transfer learning strategy increased translation accuracy by 1.8 and 1.6% for the TranSpec-CNN and TranSpec-MLP models, respectively. A hybrid approach integrating multisource and transfer learning was then developed, merging the QM9S, QM14S, and experimental spectral data sets for pretrain TranSpec, followed by transfer learning on the experimental spectra. Importantly, molecules in the test set of the experimental spectra were excluded from all of the training and validation sets to ensure data integrity. This approach improved the translation accuracy for Top 1 SMILES candidates to 19 and 11% for TranSpec-CNN and TranSpec-MLP, respectively. Model fusion was then implemented by

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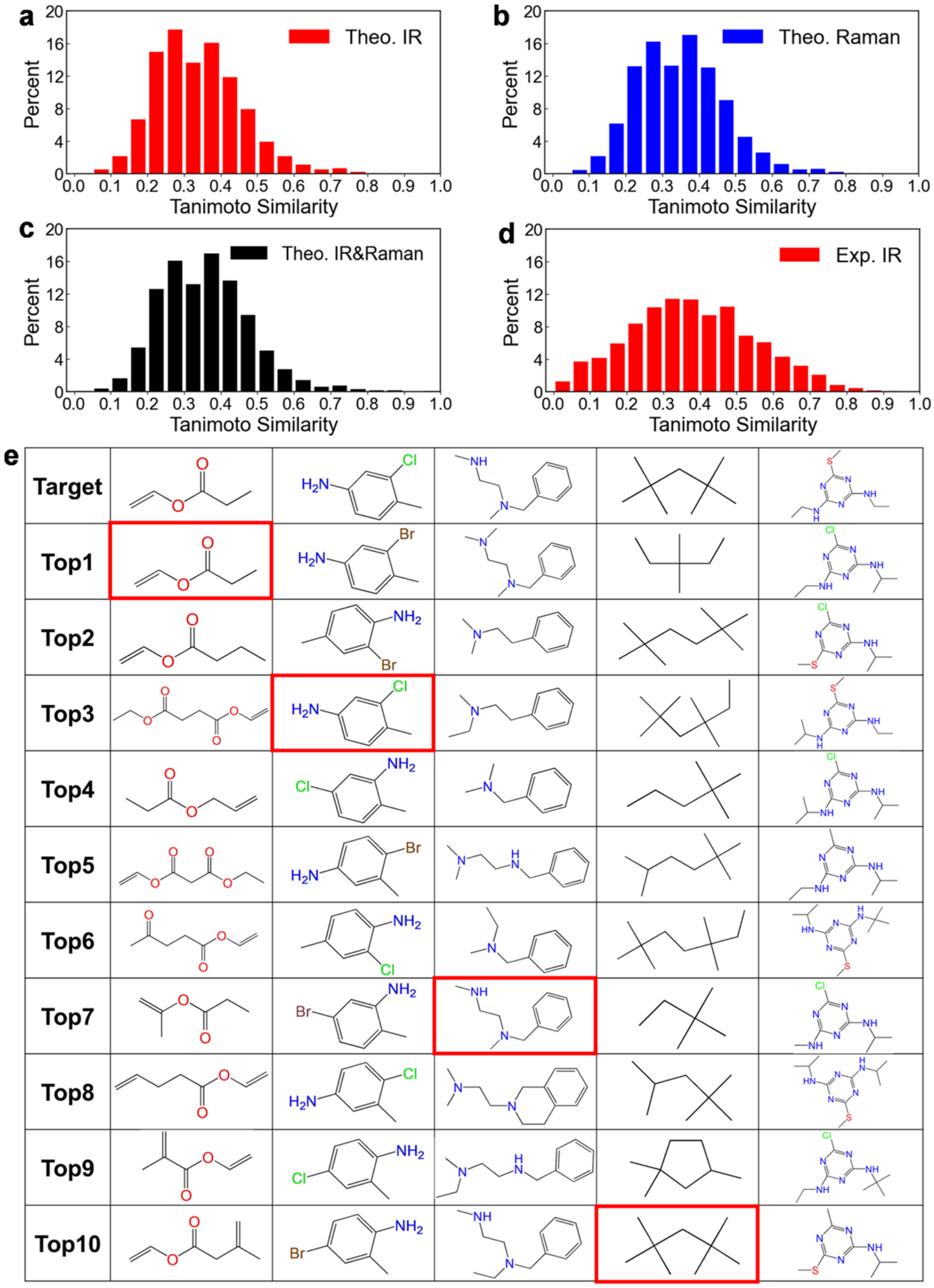

Figure 4. Tanimoto similarity distributions and examples of predicted molecules. (a–d) Histograms showing the percentage distribution of Tanimoto similarity for the Top 10 SMILES candidates across the theoretical IR, theoretical Raman, theoretical IR and Raman, and experimental IR data sets. (e) Structural comparison between the correct molecule and the Top 10 SMILES candidates for five molecules randomly selected from the experimental IR data set. Here, the red box indicates the correct translation.

adjusting learning and dropout rates to create four distinct TranSpec models (two variants each of TranSpec-CNN and TranSpec-MLP). These models were trained on the quantized, scaled, shifted, and untreated experimental data sets, yielding 16 sets of SMILES candidates. Model fusion further enhanced the translation accuracy, achieving 25.1 and 53.6% for the Top 1 and Top 10 candidates, respectively.

To further augment translation accuracy, we incorporated two methods to process the SMILES candidates. The first method involved filtering candidates based on molecular mass, where we eliminated candidates with incorrect mass values. This method increased the translation accuracy to 49.8%, with further details in Table 1. The second method focused on simulating the candidates' spectra and reordering them based on the spectral similarity calculations. Specifically, we used the Gaussian 16 package to perform frequency analysis and generate harmonic IR spectra. We constructed a ranking function based on the initial rank output from TranSpec ($Rank_{initial}$), root-mean-square error (RMSE),[60] cosine similarity (CS),[61] Pearson correlation,[62] and Spearman[63] correlation to increase the likelihood of prioritizing the correct candidates

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Table 2. Functional Group Identification Performance[a]

	Theo. IR		Theo. Raman		Theo. IR and Raman		Exp. IR	
functional group	accuracy	F1	accuracy	F1	accuracy	F1	accuracy	F1
alkane	0.962	0.996	0.976	0.997	0.988	0.999	0.561	0.899
alkene	0.974	0.987	0.992	0.992	0.994	0.995	0.708	0.800
alkyne	0.995	0.997	0.999	0.999	0.999	0.999	0.444	0.533
aromatics	0.959	0.992	0.980	0.995	0.990	0.998	0.822	0.934
alcohols	0.994	0.997	0.990	0.995	0.996	0.998	0.891	0.900
aldehydes	0.990	0.993	0.993	0.995	0.996	0.997	0.889	0.889
ketones	0.975	0.978	0.975	0.975	0.982	0.985	0.836	0.879
esters	0.972	0.953	0.970	0.965	0.977	0.975	0.862	0.871
ether	0.982	0.988	0.980	0.989	0.994	0.996	0.784	0.889
amines	0.945	0.969	0.949	0.971	0.973	0.986	0.716	0.856
nitriles	0.999	0.999	0.999	0.998	0.999	0.999	0.500	0.542
amides	0.957	0.962	0.956	0.953	0.974	0.974	0.842	0.878

[a]Identification accuracies and F1 scores obtained by TranSpec for the 12 most common functional groups across the theoretical IR, Raman, IR and Raman, and experimental IR data sets.

$$\text{Rank} = \nu_1 \text{Rank}_{\text{initial}} + \nu_2 \text{RMSE} + \nu_3 \text{CS} + \nu_4 \text{Pearson} + \nu_5 \text{Spearman} \quad (13)$$

where ν_1, ν_2, ν_3, ν_4, and ν_5 are the optimized parameters. However, we observed that reordering based on the harmonic spectral simulations did not lead to an improvement in translation accuracy.

Increasing the Accuracy of Simulated Spectra with a Graph Neural Network. A graph neural network called SpecGNN (Figure 3a) was developed to bypass the need for DFT calculations, which include electronic structural calculation, geometric optimizations, and frequency analyses. Instead, SpecGNN directly predicts experimental spectra from the SMILES representations. The process begins by converting SMILES strings into molecular 2D graphs, after which the spectra are simulated using embedded atomic and bond features. A critical consideration in this approach is ensuring that the division of training, test, and validation sets is identical across both SpecGNN and TranSpec. If the data sets are not aligned, SpecGNN may predict spectra for molecules that were part of its training set during the candidate reordering process. This would lead to an overestimation of model performance, as the model would effectively be evaluated on data it has already seen, rather than on truly independent test data. Therefore, maintaining identical data set splits is essential to ensure the integrity and generalizability of the model's predictions.

Figure 3b−d compares the SpecGNN-predicted, DFT-predicted spectra, and experimental spectra for three molecules, 2,3-dihydroxypropyl formate, 4-chlorotoluene, and 3,5-dimethoxyphenyl isothiocyanate, which were randomly selected from the test set of the experimental IR data set. The results demonstrate that SpecGNN more accurately reproduces experimental spectra compared to DFT. For example, in the case of 2,3-dihydroxypropyl formate, SpecGNN accurately predicts the peak corresponding to the C−O stretching vibrational mode at 1750 cm^{-1}, while DFT predicts a red-shifted peak at 1690 cm^{-1}, which is narrower and exhibits a weaker intensity relative to the experimental spectrum (Figure 3b). For 4-chlorotoluene, SpecGNN accurately captures the two main C−H stretching peaks near 3000 cm^{-1}, whereas DFT predicts four nondegenerate, overly intense peaks (Figure 3c). Similarly, for 3,5-dimethoxyphenyl isothiocyanate, DFT significantly underestimates the IR intensity of the aromatic ring's C═C stretching vibrations at 1600 cm^{-1} and fails to accurately predict the full width at half-maximum of the peak at 2000 cm^{-1}, both of which are well-predicted by SpecGNN (Figure 3d).

A comprehensive analysis of the entire test set was conducted to evaluate the performance of SpecGNN and DFT in simulating the spectra. Figure 3e−h shows the distributions of the RMSE and the spectral similarity metrics, comprising cosine similarity, Pearson correlation coefficient, and Spearman correlation coefficient, between the SpecGNN-predicted and DFT-predicted spectra relative to the experimental spectra. The results reveal that SpecGNN achieves significantly lower errors and higher spectral similarity to the experimental IR spectra compared to DFT.

We then replaced DFT with SpecGNN to simulate the spectra of the Top50 SMILES candidates and reordered the rankings using eq 1. It is important to note that the parameters ν_1, ν_2, ν_3, ν_4, and ν_5 required reoptimization. After reordering with SpecGNN, the final translation accuracy improved significantly, reaching 32.4% for the Top 1 candidate and 59.1% for the Top 10 candidates.

Structural Similarity Analysis of Incorrect Translations. The structural differences between the predicted and the correct molecules were inspected to assess the translation capability of our model. Using the Morgan fingerprints,[64] we calculated the Tanimoto similarity[65] between the correct SMILES and the Top 10 candidates after SpecGNN reordering. As shown in Figure 4a−d, over 60% of SMILES candidates exhibit a Tanimoto similarity greater than 0.3 compared to the query molecule for theoretical IR, Raman, IR and Raman, and experimental IR spectra. This suggests that, even in cases of incorrect translation, our model generates molecules that are structurally similar to the query. Figure 4e provides a structural comparison between the correct molecule and the Top 10 SMILES candidates for five randomly selected molecules from the experimental IR data set, further highlighting the high degree of similarity between the predicted and correct SMILES.

Applications in Identifying Functional Groups. Our model's ability to recognize the 12 most common functional groups present in the QM9S and experimental IR spectral data sets was evaluated using the Top 1 candidate obtained through SpecGNN-based reranking, excluding mass filtering. Two

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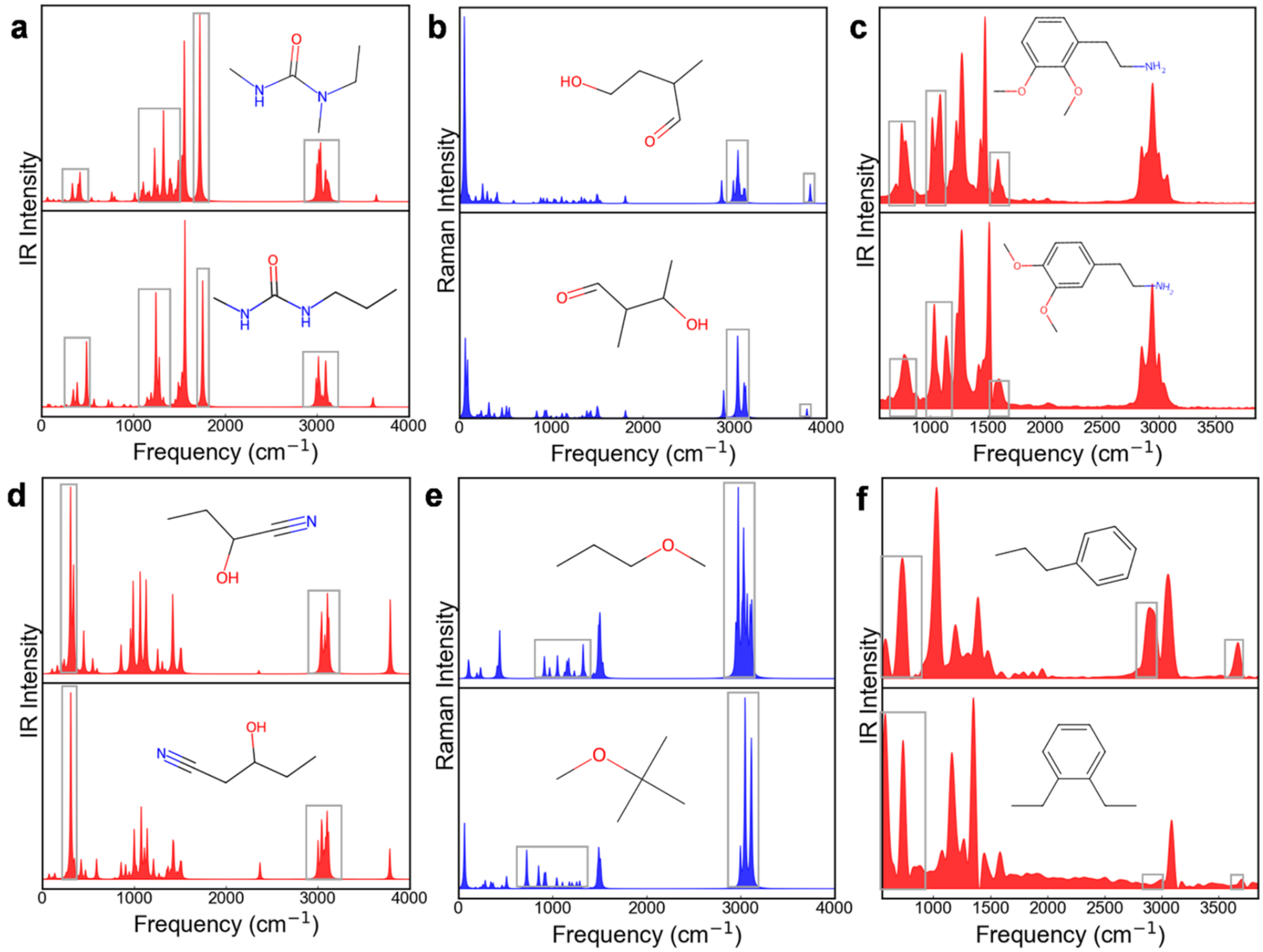

Figure 5. Application in distinguishing isomers or homologues. (a−c) Distinguishing the isomers (a) 1-ethyl-1,3-dimethylurea/1-methyl-3-propylurea, (b) 4-hydroxy-2-methylbutanal/3-hydroxy-2-methylbutanal, and (c) 2,3-dimethoxyphenethylamine/3,4-dimethoxybenzeneethanamine isomers using theoretical IR, theoretical Raman, and experimental IR spectra, respectively. (d−f) Distinguishing the (d) propionaldehyde cyanohydrin/3-hydroxyvaleronitrile, (e) methyl *n*-propyl ether/methyl *tert*-butyl ether, and (f) *n*-propylbenzene/1,2-diethylbenzene homologues using theoretical IR, theoretical Raman, and experimental IR spectra.

metrics were employed to model the performance of the model in identifying functional groups. The first one is the "identification accuracy", defined as $\frac{N_{\text{correct}}}{N_{\text{total}}}$, where N_{total} represents the total number of molecules containing the target functional groups, and N_{correct} represents the number of the molecules in which both the type and number of the corresponding groups are correctly predicted. Additionally, we calculated F1 scores based on the true positives, false positives, and true negatives to obtain more insight into the model's performance. Computational details for the identification accuracy and F1 scores are provided in the Computational Details of the Functional Groups Identification Accuracy and F1 Scores section in the SI.

As shown in Table 2, the identification accuracy and the corresponding F1 scores exceed 90% for the theoretical QM9S data set when using either IR or Raman spectra individually. Furthermore, we found that the identification of the alkane, alkene, alkyne, aromatics, aldehydes, and amine groups is more effective with Raman spectra, while alcohols, ketones, esters, ethers, amides, and nitrile groups are more accurately identified with IR spectra. These findings align with physical principles: asymmetrical geometric and electronic structures typically exhibit high IR intensity, while symmetrical structures show stronger Raman activity. Combining IR and Raman spectra further enhanced the identification accuracy for all functional groups.

Both the identification accuracy and F1 scores for the experimental IR data set are lower than those for the QM9S data set, likely due to the smaller size of the experimental data set. This issue is particularly evident for alkyne and nitrile functional groups, which only appear 118 and 280 times, respectively, in the experimental data set. Further analysis reveals low recall values for these groups (Table S4), indicating that the model struggles to accurately capture information related to triple bonds. Among the functional groups studied, alkanes exhibit lower identification accuracy but higher F1 scores, suggesting that while the model can easily recognize the presence of alkanes, it often provides an inaccurate count of the number of alkane groups in a molecule.

Applications in Distinguishing Homologues and Isomers. We selected several homologues and isomers from the test sets of the QM9S and experimental IR data sets to evaluate TranSpec's ability to distinguish between them. All

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candidates were also reranked using SpecGNN without mass filtering. As shown in Figure 5, the paired spectra for both homologues and isomers are highly similar. However, our model captures the subtle spectral differences highlighted in gray boxes in Figure 5 and correctly identifies the corresponding SMILES as the Top 1 candidate.

We examined all isomers present in the test set of the experimental IR data set and found that only 17.2% of isomers could be distinguished using the Top 1 candidate obtained after SpecGNN reordering. This low accuracy isomer in distinguishing isomers is likely due to the limited translation accuracy of TranSpec to the small experimental IR data set. To improve this, higher isomer distinction accuracy could be achieved by either expanding the experimental IR data set or incorporating readily available molecular information to filter the SMILES candidates. For instance, filtering candidates by molecular mass increased the isomer distinction accuracy to 37.2%.

4. CONCLUSIONS

In this work, we developed TranSpec and SpecGNN to achieve a bidirectional "translation" between molecular vibrational spectra and their SMILES representations. TranSpec and SpecGNN represent innovative artificial intelligence tools for identifying molecular identification and structural characterization, achieving the long-sought goal of structural inversion from spectroscopy. To enhance TranSpec's translation accuracy, we implemented several strategies including producing additional SMILES candidates, combining IR and Raman together as input, augmenting the spectra data set, fusing different models, and utilizing transfer and multisource learning. On the QM9S data set, the translation accuracy of optimized TranSpec achieved accuracies of 87.6 and 99.3% for Top 1 and Top 10 SMILES candidates, respectively. On the experimental IR data set, accuracies reached 25.1% and 53.6% for Top 1 and Top 10 SMILES candidates, respectively. To further improve the method's performance on experimental data sets, we introduced two postprocessing methods: filtering candidates by molecular mass and reordering candidates based on SpecGNN spectral simulation and similarity calculations. These methods increased Top 1 translation accuracy on the experimental IR data set to 49.8 and 32.4%, respectively. Combining both methods yielded a Top 1 accuracy of 53.6%. Additionally, we demonstrated TranSpec's capability to identify various functional groups and distinguish isomers and homologues. Looking ahead, the translation accuracy of our method could be further improved by expanding the experimental spectral data set and by incorporating additional molecular features, such as dipole moment, polarizability, dielectric constant, melting, and boiling points. These advancements would strengthen the model's generalizability and practical applicability, paving the way for broader adoption in spectroscopic analysis and molecular characterization.

■ ASSOCIATED CONTENT

Data Availability Statement

All of the data sets and training results are publicly available. The QM9S data set (including 127468 molecules), QM14S data set (including 21004 molecules), and the training results (including the isomer distinguishment, the functional group recognition) are available at 10.6084/m9.figshare.26333170. The NIST experimental IR spectra are available at https://webbook.nist.gov/chemistry. All of the codes are publicly available. The code is written using the Pytorch and RDKit libraries on python 3.9 and available at https://github.com/WeiHuQLU/TranSpec.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c05010.

Architecture of CNN and MLP, hyperparameter settings of tranSpec, hyperparameter settings of SpecGNN, computational details of the functional group identification accuracy and F1 scores, statistics of functional groups in QM9S and experimental IR data sets, isomer distinguishment accuracy across different data sets, example of isomer distinguishment using tranSpec and SpecGNN, details of the data augmentation, evaluation of training and running time, visualized difference between the original and augmented spectra, and identification accuracy of NIST IR spectra as a function of molecular size (PDF)

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Notes

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